

From Pathology to Radiology: Evaluating the Applicability of Pathology Foundation Models

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Abstract. Foundation models, large-scale models pre-trained on vast and diverse datasets, effectively capture general knowledge and demonstrate strong robustness to a wide range of downstream tasks. However, the applicability of such foundation models to different domains and modalities has not been well studied. In this work, we introduce PathRadX, a cross-modal framework to evaluate the applicability of pathology foundation models to various classification tasks in radiology images. The framework integrates modality adaptation and task-specific classification strategies for efficient and effective cross-modal adaptation. Our findings highlight the cross-modal potential of pathology pre-trained models, demonstrating their applicability to radiology imaging tasks and paving the way for developing more efficient and accurate diagnostic tools.

Keywords: Foundation models · Radiology images · Deep learning.

1 Introduction

With the recent advancements in deep learning and the increasing accessibility of large-scale models, foundation models, trained on massive datasets, have gained considerable attention [1]. These models often leverage multi-modal datasets and are primarily trained through self-supervised or supervised learning methods, capturing generalizable knowledge that enables adaptation to a variety of tasks with minimal task-specific adjustments or even in a zero-shot setting. This diminishes the need for training models from scratch for each task, facilitating their application to specific tasks with limited data and annotations. As foundation models demonstrate strong adaptability across diverse tasks and datasets, domain-specific foundation models have been widely explored. Notable examples include GPT-3 [2] and BERT [3] in natural language processing, CLIP [4] in vision-language learning, and the Segment Anything Model (SAM) [5] in vision segmentation. The emergence of these models has led to significant breakthroughs across multiple domains, including language understanding, computer vision, and multi-modal learning.

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In medical imaging, several foundation models have been developed using various imaging modalities, such as RadFM [6], a radiology foundation model aimed to integrate text input with 2D or 3D medical images, SPAD-Nets [7], trained on 55 public medical image datasets which contain both 2D and 3D images, and RETFound [8], a foundation model for abnormality detection from retinal images. Recently, there have been substantial advances in pathology foundation models, such as UNI [9], Virchow [10], and RudolphV [11]. These pathology foundation models are particularly notable due to their ability to process high-resolution images that preserve fine-grained and complex structural features. Previous studies [12] have shown that foundation models are applicable across different modalities. However, the adaptability and applicability of pathology foundation models to other medical imaging modalities remain largely unexplored.

Herein, we hypothesize that pathology foundation models possess sufficient generalization ability to be effectively adapted to other medical imaging modalities, such as radiology images. Given their ability to capture fine-grained structural and morphological features from pathology images, these models may be able to extract meaningful representations from radiology images with minimal modifications. To test this hypothesis, we propose PathRadX, a cross-modal framework that adapts a pathology foundation model to tasks specific to radiology. PathRadX involves modality adaptation and task-specific classification strategies, enabling an efficient and effective adaptation of a pathology foundation model to various radiology-specific tasks. We systematically evaluate the performance of PathRadX on three radiology datasets and compare it with three foundation models from other imaging modalities to assess its effectiveness in cross-modal medical image analysis.

Our main contributions can be summarized as below:

- We propose to adapt pathology foundation models to resolve radiology imaging tasks, leveraging their strength in analyzing fine-grained structures.
- We investigate an effective adaptation framework, referred to as PathRadX, incorporating domain adaptation and task-specific classification strategies, for pathology foundation models that requires minimal modifications and fine-tuning.
- We demonstrate that pathology foundation models exhibit strong generalization capabilities to radiology images, achieving promising results in cross-modal adaptation.

2 Methodology

PathRadX is a cross-modal framework designed to evaluate the applicability of a pathology foundation model across various tasks in radiology imaging. Given a set of radiology datasets $D = \{D_i\}_{i=1}^N$ corresponding to tasks $T = \{T_i\}_{i=1}^N$, we employ a pre-trained pathology foundation model $\mathcal{P}$ to conduct each task T_i . To fairly assess the generalization ability of $\mathcal{P}$, we freeze all its pre-trained weights. However, there are modality differences, in which pathology images are typically

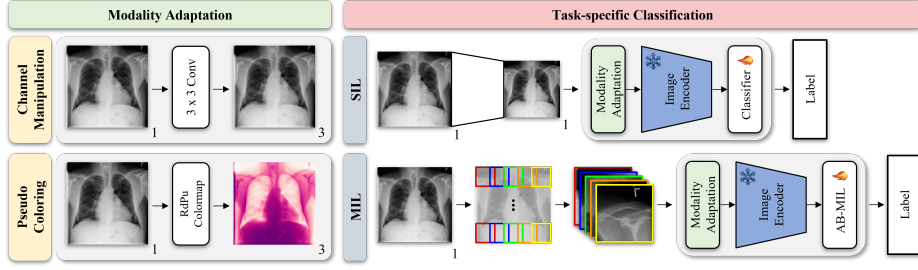


Fig. 1. Overview of the proposed PathRadX framework. To make gray-scale radiology images compatible with a pathology foundation model, we convert them into three-channel images using two different methods: pseudo-coloring with the RdPu colormap and channel manipulation via a single convolution layer. To resolve specific tasks in radiology, we employ two classification strategies: single instance learning and multiple instance learning.

three-channel color images and radiology images are generally one-channel gray-scale images, and class label differences across tasks. To address these challenges, we introduce a trainable adaptation layer $\mathcal{A}$ to bridge the modality gap and a task-specific classification layer $\mathcal{C}$ to accommodate different classification objectives. Consequently, we construct a set of pathology-based models for radiology imaging tasks, defined as $\mathcal{M} = \{\mathcal{M}_i\}_{i=1}^N$, where $\mathcal{M}_i = \mathcal{A}_i \circ \mathcal{P} \circ \mathcal{C}_i$ is tailored for each task $\mathcal{T}_i$.

2.1 Pathology Foundation Model

Among various pathology foundation models, we adopt UNI to perform radiology imaging tasks. UNI [9] is a state-of-the-art (SOTA) framework designed for pathology image analysis, which is pre-trained on Mass-100K, a large-scale dataset comprising over 100 million image patches from 100,000 whole slide images (WSIs). It is built based upon Vision Transformer Large (ViT-L) [13] architecture and utilizes DINOv2 [14] for self-supervised learning. UNI has demonstrated SOTA performance on 34 pathology image tasks, highlighting its effectiveness in capturing fine-grained structural features and complex spatial relationships in pathology images.

2.2 Modality Adaptation

Pathology foundation models are tailored to process and analyze pathology images, typically containing three-channel RGB images. UNI is not an exception. However, these models are not directly applicable to gray-scaled radiology images. To mitigate this modality discrepancy, we propose two adaptation strategies to effectively utilize pathology foundation models for radiology imaging tasks: 1) Channel manipulation; 2) Pseudo coloring.

Channel Manipulation The straightforward approach to convert a single-channel gray-scale image into a three-channel format involves the use of a convolution layer. We apply a 3×3 convolution layer with a stride of 1 and zero-padding of 1, preserving the spatial dimension of inputs while increasing the number of channels to match the expected number of channels of UNI.

Pseudo Coloring We also colorize gray-scale radiology images by mapping gray-scale intensity values into a three-channel color space. Pathology images are typically stained with hematoxylin and eosin (H&E), where cell nuclei appear purplish-blue and cytoplasm appear pink. To mimic the color patterns of H&E stained pathology images, we apply a pseudo-colorization technique using the red-purple colormap from OpenCV [15], which exhibits a color gradient from light pink/red to deep purple (Fig. 2).

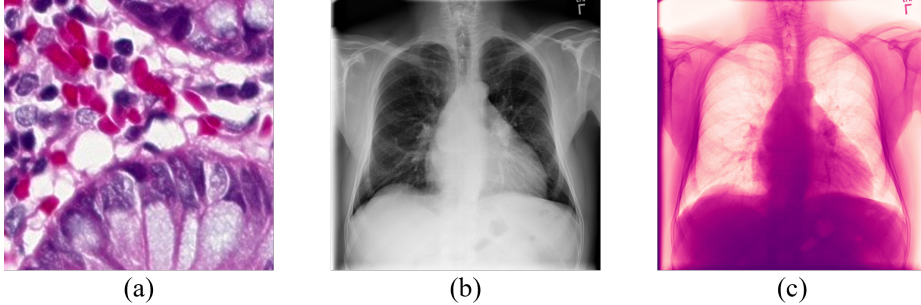


Fig. 2. Example of application of pseudo-coloring method on chest X-ray. (a) Example of a histopathology image, (b) Original chest X-ray image from the RSNA pneumonia dataset, and (c) Pseudo-colored chest X-ray image

2.3 Task-specific Classification

To adapt a pathology foundation model $\mathcal{P}$, i.e., UNI, for radiology imaging tasks, we explore two classification strategies: 1) single instance learning; 2) multiple instance learning. Given a radiology image, $\mathcal{P}$ employs one of these two classification strategies to extract a feature representation and predict the final classification label.

Single Instance Learning In this approach, we size radiology images to 224×224 pixels to match the input spatial size of UNI. The resized images are then fed into the encoder of UNI to extract a feature representation. Subsequently, the feature representation is passed through trainable linear classification layers to produce the final classification output.

Multiple Instance Learning In pathology image analysis, whole-slide images (WSIs) are extremely large, typically around $50,000 \times 50,000$ pixels, as they are acquired through high-resolution scanning. Due to their size, a direct processing of WSIs is computationally intractable. Alternatively, WSIs are divided into a number of disjoint image patches, each of which is independently processed to extract feature representations. These feature representations are then aggregated by employing multiple instance learning (MIL) approaches, generating a single representative feature representation that is used for predicting the final classification output. Recent studies have shown that pathology foundation models, equipped with MIL, achieve the SOTA performance in WSI analysis. To extend MIL to radiology images, we adopt a similar approach by cropping radiology images into multiple overlapping patches of size 224×224 pixels with an overlap of 80 pixels or 24 pixels, considering the original image size, to generate corresponding feature representations, in which each patch image is independently processed by UNI. The extracted feature representations are then aggregated using attention-based MIL (AB-MIL) [16], which utilizes adaptive attention weights to aggregate feature representations, to predict the final label.

3 Experiments

3.1 Datasets

We employ three different radiology datasets to evaluate adaptability of a pathology foundation model to radiology images. These datasets consist of X-ray images of different organs. Table 1 provides a summary of these datasets and the training, validation, and test splits for each dataset.

MURA [17] is a large dataset of musculoskeletal radiographs containing 40,561 X-ray images from 14,863 studies, containing 9,045 normal and 5,818 abnormal radiographic studies of the upper extremity including the shoulder, humerus, elbow, forearm, wrist, hand, and finger. As 556 images were used as the hidden test set during the challenge, 40,005 X-ray images were used in this study, with diverse spatial sizes up to 512×512 pixels.

RSNA Pneumonia Dataset (RSNA_P) [18] was originally made in public during RSNA Pneumonia Detection Challenge held in 2018, containing 26,686 images with the spatial size of 1024×1024 pixels each. We use this dataset for a binary classification task to determine the presence of pneumonia.

MedFMC (MedFMC_C) [19] contains 22,349 images in total, covering 5 different modalities, X-ray, pathology, endoscopy, digital camera, and retinography. Among them, we use 4,848 Chest X-ray images to conduct thoracic abnormality classification. Each image has a spatial size of 1024×1024 pixels.

Table 1. Overview of radiology datasets with detailed characteristics. Unlike other datasets, MURA has variable size of images.

Dataset	Organ	Image size	Train	Validation	Test	Total
MURA	musculoskeletal X-ray	$\leq 512 \times 512$	36,808	1,500	1,697	40,005
RSNA _P	chest X-ray	$1,024 \times 1,024$	21,348	2,668	2,668	26,684
MedFMC _C	chest X-ray	$1,024 \times 1,024$	3,878	485	485	4,848

3.2 Comparative Experiments

To evaluate the effectiveness of the proposed PathRadX framework, we conduct comparative experiments using three different radiology foundation models: 1) SAM-Med2D; 2) MedCLIP; 3) BiomedCLIP. Following the standard practice in radiology foundation models, each model processes an input radiology image through its image encoder to extract a feature representation, which is fed into trainable linear layers for classification.

SAM-Med2D SAM has set a benchmark for segmenting instances in natural images through its prompt-based approach. As a variant of SAM, SAM-Med2D [20] was built by fine-tuning SAM with approximately 4.6 million medical images and 19.7 million corresponding masks. It has been shown that SAM-Med2D outperforms existing segmentation methods on nine MICCAI2023 datasets.

MedCLIP MedCLIP [22] is a vision-language model designed for medical image analysis, pre-trained on a large dataset of chest X-ray images paired with clinical text, such as CheXpert [23] and MIMIC-CXR [24]. By learning to associate visual features with medical terminology, it has demonstrated strong performance, particularly in chest X-ray image analysis.

BiomedCLIP BiomedCLIP extends MedCLIP by focusing on broader biomedical applications. It utilizes PubMedBERT [25] as a text encoder and a ViT model as an image encoder. Pre-trained on a large biomedical dataset called PMC-15M [26], including histopathology images, radiology images, and clinical notes, BiomedCLIP enhances the interaction between biomedical images and text, improving performance across various medical imaging tasks.

3.3 Implementation and Training Details

The image encoders of each foundation model were frozen, and only the classifier weights were updated during training. Each model was trained for 100 epochs with OneCycleLR [27] scheduler (initial learning rate 0.001) and the Adam optimizer, with 0 weight decay and beta values (β_1 , β_2) set to 0.9 and 0.999. All experiments were conducted on a single NVIDIA RTX A6000 GPU. To evaluate and train the models, we adopt binary cross-entropy loss.

4 Results and Discussions

Table 2. Results of three radiology imaging tasks. CM: channel manipulation, UNI: a pathology foundation model; MIL: multiple instance learning.

Models	MURA			RSNA _P			MedFMC _C		
	Acc(%)	F1	AUC	Acc(%)	F1	AUC	Acc(%)	F1	AUC
PathRadX (CM ◦ UNI ◦ MIL)	77.31	0.777	0.861	83.70	0.576	0.863	72.99	0.812	0.768
SAM-Med2D	70.94	0.726	0.788	79.72	0.333	0.809	74.23	0.852	0.757
MedCLIP	74.96	0.696	0.844	81.90	0.488	0.834	76.29	0.848	0.784
BiomedCLIP	75.25	0.700	0.835	81.52	0.417	0.820	75.26	0.851	0.762

Table 3. Effect of different modality adaptation and task-specific classification strategies on the performance of a pathology foundation model. SIL: single instance learning, MIL: multiple instance learning, CM: channel manipulation, PC: and pseudo-coloring.

Strategy				MURA			RSNA _P			MedFMC _C		
SIL	MIL	CM	PC	Acc(%)	F1	AUC	Acc(%)	F1	AUC	Acc(%)	F1	AUC
✓		✓		77.84	0.752	0.848	81.33	0.357	0.836	79.18	0.870	0.819
✓			✓	77.61	0.744	0.849	80.88	0.348	0.844	79.59	0.873	0.807
	✓		✓	77.31	0.777	0.861	83.70	0.576	0.863	72.99	0.812	0.768
	✓		✓	77.67	0.753	0.839	81.30	0.540	0.829	77.73	0.852	0.781

Table 2 demonstrates the performance of each foundation model on radiology imaging datasets based on three metrics: accuracy (Acc), F1 score, and AUC score. Leveraging UNI with AB-MIL and channel-manipulation, we achieved the highest scores across all evaluation metrics for MURA and RSNA_P, outperforming others by $\leq 6.4\%$ Acc, ≤ 0.081 F1, and ≤ 0.073 AUC for MURA and $\leq 4.0\%$ Acc, ≤ 0.243 F1, and ≤ 0.054 AUC for RSNA_P. Meanwhile, on MedFMC_C, MedCLIP obtained the highest Acc of 76.29% and AUC of 0.784 and SAM-Med2D obtained the best F1 of 0.852. The proposed method was inferior to these models by 3.3% Acc, 0.040 F1, and 0.016 AUC.

To provide an in-depth understanding of the classification results, we computed and examined confusion matrices for each model (Figure 3). In a close-up investigation of these confusion matrices, we found that two of the three datasets (RSNA_P and MedFMC_C) were highly imbalanced. The imbalance particularly affects the performance on MedFMC_C. SAM-Med2D predicted all samples as positive. MedCLIP had much lower specificity than the proposed method, leading to a high rate of false positives. Overall, the proposed method generally demonstrated more balanced and reliable classification performance.

Moreover, we evaluated the proposed method with differing strategies for modality adaptation and task-specific classification (Table 3). Except MedFMC_C,

	UNI		SAM-Med2D		MedCLIP		BiomedCLIP			
MURA	Predicted	Positive	671	221	654	312	486	76	490	75
		Negative	164	641	181	550	349	786	345	787
	Ground Truth	Positive								
		Negative								
RSNA _p	Predicted	Positive	296	130	135	75	230	112	176	68
		Negative	305	1937	466	1992	371	1955	425	1999
	Ground Truth	Positive								
		Negative								
MedFMC _C	Predicted	Positive	283	54	360	125	321	76	343	103
		Negative	77	71	0	0	39	49	17	22
	Ground Truth	Positive								
		Negative								

Fig. 3. Visualization of confusion matrices of evaluation results.

the combination of AB-MIL and channel-manipulation provided high performance across three metrics. On MedFMC_C, equipped with SIL, the proposed method achieved much higher scores for both modality adaptation strategies, even outperforming MedCLIP. These results suggest that optimizing modality adaptation and task-specific classification strategies could further improve the performance.

Furthermore, future studies could systematically evaluate the potential of pathology foundation models on radiology tasks through broader comparisons. As an extension of our current findings, evaluating the proposed framework against models pretrained on large-scale natural image datasets may help further validate the robustness of PathRadX for radiology imaging tasks. Additionally, incorporating models such as CTransPath [28] or Lunit [29] into follow-up experiments would enable exploration of cross-domain strategies, potentially improving the effectiveness of pathology-specific models when applied to radiology tasks.

5 Conclusions

In this paper, we investigate four different strategies within the PathRadX framework to leverage a pathology foundation model (UNI) on radiology imaging tasks in comparison to radiology foundation models. The results suggest that pathol-

ogy foundation models are promising to conduct various radiology imaging tasks. However, the specific strategies for adaptation needs further investigation. In a follow-up study, we aim to explore more effective strategies to adapt pathology foundation models to various medical imaging tasks and assess the adaptability of this approach across different domains and tasks in medical image analysis.

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References

1. Bommasani R, Hudson DA, Adeli E, Altman R, Arora S, von Arx S, Bernstein MS, Bohg J, Bosselut A, Brunskill E, Brynjolfsson E.: On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258 (2021)
2. Brown T, Mann B, Ryder N, Subbiah M, Kaplan JD, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A, Agarwal S.: Language models are few-shot learners. *Advances in neural information processing systems*, **33**, pp. 1877-901 (2020)
3. Devlin J, Chang MW, Lee K, Toutanova K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171-4186 (2019)
4. Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, Sastry G, Askell A, Mishkin P, Clark J, Krueger G.: Learning transferable visual models from natural language supervision. In: *International conference on machine learning*, pp. 8748-8763 (2021)
5. Kirillov A, Mintun E, Ravi N, Mao H, Rolland C, Gustafson L, Xiao T, Whitehead S, Berg AC, Lo WY, Dollár P.: Segment anything. In: *Proceedings of the IEEE/CVF international conference on computer vision 2023*, pp. 4015-4026 (2023)
6. Wu C, Zhang X, Zhang Y, Wang Y, Xie W.: Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. arXiv preprint arXiv:2308.02463 (2023)
7. Luo L, Chen X, Tang B, Chen X, Han R, Hu C, Li Y, Chen T.: Building Universal Foundation Models for Medical Image Analysis with Spatially Adaptive Networks. arXiv preprint arXiv:2312.07630 (2023)
8. Zhou, Y., Chia, M.A., Wagner, S.K. et al.: A foundation model for generalizable disease detection from retinal images. *Nature* **622**, pp. 156–163 (2023). <https://doi.org/10.1038/s41586-023-06555-x>
9. Chen RJ, Ding T, Lu MY, Williamson DF, Jaume G, Chen B, Zhang A, Shao D, Song AH, Shaban M, Williams M.: A general-purpose self-supervised model for computational pathology. arXiv preprint arXiv:2308.15474 (2023)
10. Vorontsov E, Bozkurt A, Casson A, Shaikovski G, Zelechowski M, Liu S, Severson K, Zimmermann E, Hall J, Tenenholtz N, Fusi N.: Virchow: A million-slide digital pathology foundation model. arXiv preprint arXiv:2309.07778 (2023)

11. Dippel J, Feulner B, Winterhoff T, Milbich T, Tietz S, Schallenberg S, Dernbach G, Kuntz A, Heinke S, Eich ML, Ribbat-Idel J.: RudolfV: a foundation model by pathologists for pathologists. arXiv preprint arXiv:2401.04079 (2024)
12. Huix JP, Ganeshan AR, Haslum JF, Söderberg M, Matsoukas C, Smith K.: Are natural domain foundation models useful for medical image classification?. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision 2024, pp. 7634-7643 (2024)
13. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, Uszkoreit J.: An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929 (2020)
14. Oquab M, Darcet T, Moutakanni T, Vo H, Szafraniec M, Khalidov V, Fernandez P, Haziza D, Massa F, El-Nouby A, Assran M.: Dinov2: Learning robust visual features without supervision. arXiv preprint arXiv:2304.07193 (2023)
15. Bradski, G., Kaehler, A.: OpenCV: Open Source Computer Vision Library. In: Proceedings of the Intel Technology Journal, vol. 2000, pp. 1-11. <https://opencv.org>
16. Ilse M, Tomczak J, Welling M.: Attention-based deep multiple instance learning. In: International conference on machine learning 2018, pp. 2127-2136 (2018)
17. Rajpurkar P, Irvin J, Bagul A, Ding D, Duan T, Mehta H, Yang B, Zhu K, Laird D, Ball RL, Langlotz C.: Mura: Large dataset for abnormality detection in musculoskeletal radiographs. arXiv preprint arXiv:1712.06957 (2017)
18. RSNA Pneumonia Detection Challenge, <https://kaggle.com/competitions/rsna-pneumonia-detection-challenge>
19. Wang D, Wang X, Wang L, Li M, Da Q, Liu X, Gao X, Shen J, He J, Shen T, Duan Q.: A real-world dataset and benchmark for foundation model adaptation in medical image classification. Scientific Data, 2023, **10**(1), p.574 (2023)
20. Cheng J, Ye J, Deng Z, Chen J, Li T, Wang H, Su Y, Huang Z, Chen J, Jiang L, Sun H.: Sam-med2d. arXiv preprint arXiv:2308.16184 (2023)
21. Wang X, Peng Y, Lu L, Lu Z, Bagheri M, Summers RM.: Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 2097-2106 (2017)
22. Wang Z, Wu Z, Agarwal D, Sun J.: Medclip: Contrastive learning from unpaired medical images and text. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing, vol. 2022, p. 3876 (2022)
23. Irvin J, Rajpurkar P, Ko M, Yu Y, Ciurea-Ilcus S, Chute C, Marklund H, Haghighi B, Ball R, Shpanskaya K, Seekins J.: Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In: Proceedings of the AAAI conference on artificial intelligence, vol. 33, No. 01, pp. 590-597 (2019)
24. Johnson AE, Pollard TJ, Greenbaum NR, Lungren MP, Deng CY, Peng Y, Lu Z, Mark RG, Berkowitz SJ, Horng S.: MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs. arXiv preprint arXiv:1901.07042 (2019)
25. Gu Y, Tinn R, Cheng H, Lucas M, Usuyama N, Liu X, Naumann T, Gao J, Poon H.: Domain-specific language model pretraining for biomedical natural language processing. ACM Transactions on Computing for Healthcare (HEALTH), **3**(1), pp. 1-23 (2021)
26. Zhang S, Xu Y, Usuyama N, Xu H, Bagga J, Tinn R, Preston S, Rao R, Wei M, Valuri N, Wong C.: Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. arXiv preprint arXiv:2303.00915 (2023)

27. Smith LN, Topin N.: Super-convergence: Very fast training of neural networks using large learning rates. In: Artificial intelligence and machine learning for multi-domain operations applications, vol. 11006, pp. 369-386 (2019)
28. Xiyue W, Sen Y, Jun Z, Minghui W, Jing Z, Wei Y, Junzhou H, Xiao H.: Transformer-based unsupervised contrastive learning for histopathological image classification. In: Medical image analysis, vol. 81 (2022)
29. Mingu K, Heon S, Seonwook P, Donggeun Y, Sérgio P.: Benchmarking self-supervised learning on diverse pathology datasets. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3344-3354 (2023)